

# Quantum Fault Tolerance Meets Toom's Contours

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## 1 Computation Model

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Separate between an element gate  $g \in \mathcal{G}$  and a gate, which also has a space-time location.

**Definition 1.1** (Quantum Circuit). A space-time location  $l$  is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates  $\mathcal{G} = \{g: L(\mathcal{H}_2^{\otimes \Delta}) \rightarrow L(\mathcal{H}_2^{\otimes \Delta})\}$ , and integers  $w, d \in \mathbb{N}$  a quantum circuit  $C$ , with depth  $d$  and width  $w$ , is a collection of tuples  $\{u_i = (g_i, l_i)\}_{i=1}$ , where  $u_i$  are the gates of  $C$  and each  $u_i$  is composed of a gate  $g_i \in \mathcal{G}$  and a space-time location  $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^\Delta$ . The gates of  $C$  have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for  $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$ , the equality  $l_{i,1} = l_{j,1}$  implies  $l_{i,2} \cap l_{j,2} = \emptyset$ . We define the  $t$ -cut of  $C$  to be  $C|_t := \{u_i = (g_i, l_i): u_i \in C \text{ and } l_{i,1} = t\}$ .

To define the running of a quantum circuit  $C$  on a given (mixed) quantum state, we will need to have the notion of the directed acyclic graph associated with  $C$ .

**Definition 1.2** (Associated Computation DAG). Let  $C = \{u_i\}_{i=1}$  be a quantum circuit with depth  $d$  and width  $w$ . We define the associated computation DAG of  $C$ , denoted by  $\pi_C$ , to be the acyclic directed graph  $\pi_C = (\mathcal{U}, E)$  as follows:

1. The vertex set  $\mathcal{U}$  is the gate set of  $C$ , namely  $\mathcal{U} = \{u_i\}_{i=1}$ .
2. For  $u_i = (g_i, l_i), u_j = (g_j, l_j)$ , There is an edge  $\{u_i, u_j\} \in E$  if and only if:

$$j \in \arg \min_k \{l_{k,1} : (g_k, l_k) \in \mathcal{U}, l_{k,1} > l_{i,1} \text{ and } l_{k,2} \cap l_{i,2} \neq \emptyset\}$$

Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that  $J = \{j_1, j_2, \dots, j_k\} \subset [|C|]$  is an order of the indices of the gates in  $C$  according to the topological order induced by  $\pi_C$ . Then, we say that  $J$  is an order of indices that agrees with  $\pi_C$  and define the series of circuit  $C_J = \{C_{J,i}\}_{i=1}$  as follows:

1. The first element:  $C_{J,1} = \{u_{j_1}\}$
2. Then,  $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$ . We name  $C_{J,i}$  the partial computation circuit until step  $i$ .

**Definition 1.3.** Let  $\rho$  be a density matrix over  $w$  qubits, and let  $C = \{u_i\}_{i=1}$  be a quantum circuit with depth  $d$  and width  $w$ . For any density matrix  $\sigma$ , quantum gate  $g \in \mathcal{G}$ , and location  $l$ , the application of the gate-location tuple  $(g, l)$  on  $\sigma$ , denoted by  $(g, l) \circ \sigma$ , is the application of  $g$  on  $\sigma$  where wires are ordered such that the first qubit in the input to  $g$  is  $l_{2,1}$ , the second is  $l_{2,2}$ , and so on. The result of the application of  $C$  on  $\rho$ , denoted by  $C \circ \rho$ , is given by iterative applications of the gates  $\{u_i\}_{i=1}$  on  $\rho$ , according to a topological order induced by  $\pi_C$ .

We say that  $\rho_1, \rho_2, \dots, \rho_k$  is a computation prefix if  $\rho_1 = \rho$  and  $\rho_{i+1} = u'_i \circ \rho_i$ , where  $\{u'_i\}_{i=1}$  is a prefix of a topological ordering of  $\{u_i\}_{i=1}$ . If  $\{u'_i\}_{i=1}$  contains all the gates in  $\{u_i\}_{i=1}$ , then we call  $\rho_1, \rho_2, \dots, \rho_k$  a running.

**Definition 1.4** ( $C$  Subjected to  $\mathcal{E}$  at frequency  $r$ ). Let  $C = \{u_i\}_{i=1}$  be a quantum circuit with depth  $d$  and width  $w$ . For a set of fault locations  $\mathcal{E} \subset [d] \times [w]$ , and a number  $r \in (0, 1)$  satisfies  $r^{-1} \in \mathbb{N}$ , we define the quantum circuit  $C$  subjected to  $\mathcal{E}$  at frequency  $r$ , denoted by  $C[\mathcal{E}]_r = \{u'_i\}_{i=1}$ , to be the quantum circuit that is the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) = \begin{cases} (g_i, (l_{i,1} + \lfloor r l_{i,1} \rfloor + 1, l_{i,2})) & u_i \in C \\ (g_i, (r^{-1} l_{i,1}, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

Where  $r$  is not mentioned, we mean  $r = \frac{1}{2}$ , and the result of the subsection is the circuit whose gates on the odd time coordinates are the gates of  $C$ , and whose gates on the even time coordinates are the gates of  $\mathcal{E}$ . We will denote the associated computation DAG of  $C$  affected by  $\mathcal{E}$  as  $\pi[C, \mathcal{E}]$ .

**Definition 1.5** (Faults Propagation to a Step  $\tau$ ). For a quantum circuit  $C = \{u_i\}_{i=1}$ , with depth  $d$ , an order  $J = \{j_1, j_2, \dots, j_k\} \subset [|C|]$  that agrees with  $\pi_C$ , and gate  $u_i$  in  $C$ , a propagation of a gate  $u_i \in C$  to step  $\tau$  is the quantum operator  $X: L(\mathcal{H}_2^w) \rightarrow L(\mathcal{H}_2^w)$  obtained by the follow process:

1. First, check if  $i$  precedes  $\tau$  in  $J$ . If not, then return  $X = I$ . Otherwise, denote  $j_l = i$ , and look for an operator  $X_1$  such that  $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$ .
2. Then, for  $k \leq \tau - l$ , look for an operator  $X_k: L(\mathcal{H}_2^w) \rightarrow L(\mathcal{H}_2^w)$ , such that  $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$ .
3. At the final stage of the process, namely where  $k = \tau - l$ , we define  $X := X_k$  to be the propagation of  $u_j$  to step  $\tau$ .

We define the propagated computation graph  $(\mathcal{U}', E')$ , resulted by the propagation of  $u_i$  to step  $\tau$ , as follow: Let  $u_i = u_{j_1}, \dots, u_{j_l}$ , be the gates at steps between  $i$  to  $\tau$ , and let  $X$  be the propagation of  $u_i$  to step  $\tau$ . Consider the tree rooted at  $u_i$  restricted to  $u_{j_1}, \dots, u_{j_l}$ , denote it by  $T$  and it leaves by  $L_T$ . Set  $\mathcal{U}' = \mathcal{U}/u_i \cup X$  and:

$$\begin{aligned} E' = & E \\ & / \{ \{u, u_i\} : \{u, u_i\} \in E \} \\ & / \{ \{u, v\} : u \in L_T, \{u, v\} \in E \} \\ & \cup \{ \{X, v\} : u \in L_T, \{u, v\} \in E \} \\ & \cup \{ \{u, X\} : u \in L_T \} \\ & \cup \{ \{u, v\} : \{u, u_i\}, \{u_i, v\} \in E \} \end{aligned}$$

For a quantum circuit  $C$ , a set of faults  $\mathcal{E} = \{e_i\}_{i=1}$ , an order  $J \subset [|C[\mathcal{E}]|]$  that agrees with  $\pi_{C[\mathcal{E}]}$ , and a step  $\tau \in J$ , denote by  $X_{e_i}^\tau$  the propagation of  $e_i \in \mathcal{E}$  in  $C[\mathcal{E}]$  to step  $\tau$ . Let  $j'_1, j'_2, \dots, j'_l$  be the indices of the faults in  $\mathcal{E}$  such that their corresponding positions in  $J$  precede  $\tau$ , ordered according to the reverse topological order induced by  $\pi_{\mathcal{E}}$ . The propagated error to step  $\tau$  is defined by

$$X^\tau := \prod_i X_{e_{j'_i}}^\tau$$

**Definition 1.6** ( $C \sim_P C'$ ). We define the equivalence relation over quantum circuits  $\sim_P$  to be satisfied if any of them can be obtained from the other by propagating gates. Namely, for quantum circuits  $C, C'$ , we have  $C \sim_P C'$  if there is a subset of gates  $\{u_i\}_{i=1} \subset C$  and subsets of turns  $\{\tau_i\}_{i=1}$  such that  $C'$  is the result of propagating  $u_i$  to turn  $\tau_i$  in  $C$  according to the order indexed by  $i$ .

Note that, by definition, we find that equivalence between quantum circuits by  $\sim_P$  implies an equivalence of their action. More specifically, for a mixed state  $\rho$  and quantum circuits  $C$  and  $C'$  such that  $C \sim_P C'$ , we have:

$$C\rho C^\dagger = C'\rho C'^\dagger$$

**Definition 1.7** (Deviation). For a quantum circuit  $C$ , a set of faults  $\mathcal{E}$ , and step  $\tau$ , let  $X^\tau$  be the propagated error at step  $\tau$ . We define the deviation of  $C[\mathcal{E}]$  from  $C$  at step  $\tau$  by the set of indices on which  $X^\tau$  acts non-trivially.

**Definition 1.8.** Using the same notations as in Definition 1.7, for a set of unitary operators  $\mathcal{S} = \{s_i : \mathcal{H}_2^w \rightarrow \mathcal{H}_2^w\}_{i=1}$ , we define the propagated error up to stabilizers in  $\mathcal{S}$  as  $X^\tau[S] := \min_{s \in \mathcal{S}} \{w(sX)\}$ . Then, we define the deviation of  $C[\mathcal{E}]$  from  $C$  at step  $\tau$  up to stabilizers in  $\mathcal{S}$  to be the subset of coordinates on which  $X^\tau[S]$  acts non-trivially.

The following, extends the notion of Tanner graphs to the context of quantum circuits.

**Definition 1.9** (Tanner Graph of a Register). For a quantum circuit  $C$ , a register  $R$  is a subset of qubits in  $C$ . The *Tanner graph* at step  $\tau$  of register  $R$ , denoted by  $\mathcal{T}_{R,\tau}$ , is the graph defined as follows:

1. If there is a code  $\mathcal{C}$ , such that at turn  $\tau$ , the qubits in  $C$  are an encoding of  $\mathcal{C}$ , then  $\mathcal{T}_{R,\tau}$  is the Tanner graph of  $\mathcal{C}$ .
2. Otherwise,  $\mathcal{T}_{R,\tau}$  is the empty graph.

Where we consider a range of turns in which  $R$  is encoded in the same code, and the turn is clear from the context, we will be brief by omitting the turn subscript.

**Definition 1.10** (Generalized Tanner Graph of a Circuit). For a quantum circuit  $C$ , and registers  $a$  and  $b$  in  $C$ , the *generalized Tanner graph* of the register obtained by taking their union  $a \cup b$  is  $\mathcal{T}_{a \cup b} := \mathcal{T}_a \cup \mathcal{T}_b$ . For a partition of the qubits into registers  $\{R_i\}_{i=1}$ , the *generalized Tanner graph* of a quantum circuit  $C$ , at step  $\tau$ , according to the partition  $\{R_i\}_{i=1}$ , is the graph obtained by:

$$\mathcal{T}_{C,\{R_i\}_{i=1},\tau} := \bigcup_{i=1} \mathcal{T}_{R_i,\tau}$$

When the partition is clear, we will brief and use  $\mathcal{T}_C$  to define the generalized Tanner graph of  $C$ .

**Definition 1.11** (Clustered Deviation). For a quantum circuit  $C$ , a register  $R$  of  $C$ , and a set of faults  $\mathcal{E}$ , let  $\mathcal{T}_R$  be the Tanner graph of  $R$ , and let  $\mathcal{D}_\tau$  be the deviation of  $C[\mathcal{E}]$  from  $C$  on  $R$  at step  $\tau$ . Denote by  $\mathcal{T}_R|_{\mathcal{D}_\tau}$  the subgraph of  $\mathcal{T}_R$  induced by the restriction of the left vertices (bits) to the indices in  $\mathcal{D}_\tau$ . Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step  $\tau$  to be the connected components of  $\mathcal{T}_R|_{\mathcal{D}_\tau}$ . The size of each deviation cluster is the number of right vertices it contains.

**Claim 1.12.** Let  $C$  be a quantum circuit composed of gates with bounded fan degree  $\Delta \in \mathbb{N}$ . In addition, let  $R$  be a quantum register encoded by a quantum error correction code  $\mathcal{C}$  with Tanner graph  $\mathcal{T}$  that has left and right degrees  $\Delta_1$  and  $\Delta_2$ . Fix a set of faults  $\mathcal{E}$ , and an order  $J$  that agrees with  $\pi_{C[\mathcal{E}]}$ . For any  $\tau$ , denote by  $Y_1^{(\tau)}, Y_2^{(\tau)}, \dots, Y_l^{(\tau)}$  the deviation clusters at step  $\tau \in J$ . Assume that at step  $\tau_1 \in J$  there is a deviation cluster  $Y_i^{(\tau_1)}$  of size  $|Y_i^{(\tau_1)}|$ . Then there is a deviation cluster at step  $\tau_1 - 1$ , let's denote it by  $Z$ , such  $Z \cap Y_i^{(\tau_1)} \neq \emptyset$  and  $Z$  is of size at least:

$$|Z| \geq \frac{1}{\Delta \Delta_1 \Delta_2} |Y_i^{(\tau_1)}|$$

Furthermore there is a deviation cluster at step  $\tau_1 - 1$ , let's denote it by  $Z'$ , such  $Z' \cap Y_i^{(\tau_1)} \neq \emptyset$ , and  $Z'$  is of size at most:

$$|Z'| \leq \Delta \Delta_1 \Delta_2 |Y_i^{(\tau_1)}|$$

Add: that when it isn't mentioned, we assume that the error at time  $t$  can be decomposed into a product  $\Rightarrow$  at each step, a fault is supported on at most one (q)bit.

*Proof.* Denote by  $u = (g, l)$  the gate at step  $\tau$ . If  $u$  acts trivially on the (q)bits in  $Y_i^{(\tau_1)}$  then they also belong to deviation on register  $R$  at step  $\tau_1 - 1$ . Since they are connected in  $\mathcal{T}$  they form a deviation cluster on register  $R$  at step  $\tau_1 - 1$  and we got the desired. Thus it is left to handle the case in which  $u$  acts non-trivially on (q)bits in  $Y_i^{(\tau_1)}$ . Since  $u$  acts on at most  $\Delta$  qubits, and each qubit has at most  $\Delta_1(\Delta_2 - 1)$  neighbors at  $\mathcal{T}$  it follows that  $u$  acts non trivially on at most

$$\Delta + \Delta\Delta_1(\Delta_2 - 1) \leq \Delta\Delta_1\Delta_2$$

deviation clusters at step  $\tau_1 - 1$ . Let  $l$  be integer no greater than  $\Delta\Delta_1\Delta_2$  and let  $Y_1, Y_2, \dots, Y_l$  be the deviation cluster, on register  $R$ , at step  $\tau_1 - 1$ , on which  $u$  acts non trivially, since  $Y_i^{(\tau_1)} \subset \bigcup_j Y_j$ . We have that:

$$\begin{aligned} |Y_i^{(\tau_1)}| &\leq \sum_j |Y_j| \leq \Delta\Delta_1\Delta_2 \max_j |Y_j| \\ \Rightarrow \max_j |Y_j| &\geq \frac{1}{\Delta\Delta_1\Delta_2} |Y_i^{(\tau_1)}| \end{aligned} \tag{1}$$

**Complete the second inequality.** Should consider to erase what have already written about it. To get the second inequality, we use the fact that  $u$  is a unitary gate, in particular,  $u$  is reversible. Let  $C_{J, \tau_1} = u_1, \dots, u_{\tau_1}$  be the partial computation circuit until step  $\tau_1$ . Observe that  $C_{J, \tau_1 - 1}$  is obtained by stacking  $u_{\tau_1}^\dagger$  as a final stage to  $C_{J, \tau_1}$ , denoted by  $u_{\tau_1}^\dagger \circ C_{J, \tau_1}$ . Thus, the deviation from  $C_{J, \tau_1}$  at step  $\tau_1 - 1$  equals the deviation from  $u_{\tau_1}^\dagger \circ C_{J, \tau_1}$  at step  $\tau_1 + 1$ . By eq. (1),  $\square$

**Corollary 1.13** (Intermediate Value Theorem). Let  $\alpha = 1/(\Delta\Delta_1\Delta_2)$ , and let  $x$  be an integer. Assume that at the initial step  $\tau_o$ , there is no deviation cluster of size more than  $\alpha x$ , and that at a step  $\tau_1 \in J$ , such that  $\tau_1 > \tau_o$ , there is a deviation cluster  $Y^{(\tau_1)}$  of size  $|Y^{(\tau_1)}| > x$ . Then there is a step  $\tau_\star \in J$  such that  $\tau_o < \tau_\star < \tau_1$  for which there is a deviation cluster  $Z$  at step  $\tau_\star$ , satisfying that the size of  $Z$  is in the range  $(\alpha x, x)$ .

*Proof.* Assume through contradiction that for any turn  $\tau \in (\tau_o, \tau_1)$ , all the deviation clusters are of size smaller than  $\alpha x$ . Particularly, at turn  $\tau_1 - 1$ , all the deviation clusters are of size smaller than  $\alpha x$ . Yet, by Claim 1.12, the existence of the deviation cluster  $Y^{(\tau_1)}$ , at size  $d$ , implies the existence of a deviation cluster at turn  $\tau_1 - 1$ , of size at least  $\alpha x$ , which leads to a contradiction.  $\square$

**Claim 1.14.** Let  $C$  be a quantum circuit with order  $J$  and depth  $d$ , and let  $\mathcal{E} \sim \mathcal{N}_{\mathcal{E}}$ , so the circuit  $C[\mathcal{E}]$  is a random variable, and it is fan-bounded by  $\Delta$ . For a time  $t \leq 2d$  and an integer  $x \in \mathbb{N}$ , let  $A$  be the event in which, until time  $t$ , there is a time  $t' \leq t$  such that there is a deviation cluster of size greater than  $x$ . For any step  $\tau \in J$ , let  $B_\tau$  be the event that step  $\tau$  is the first step to have a deviation cluster of size in the range  $(\alpha x, x)$ . Then:

$$\Pr[A] \leq 2wd \max_{\tau \in [2wd]} \Pr[B_\tau]$$

*Proof.* By Corollary 1.13, we have that the occurrence of a deviation cluster of size greater than  $x$  before time  $t$  implies that there is a step  $\tau \in J$ , such that  $\tau \leq 2wt$ , in which there is a deviation cluster of size greater than  $\alpha x$ . Thus,

$$A \subset \bigcup_{\tau \in [2wt]} B_\tau \subset \bigcup_{\tau \in [2wd]} B_\tau$$

So by using the union bound we get:

$$\begin{aligned} \Pr[A] &\leq \Pr \left[ \bigcup_{\tau \in [2wd]} B_\tau \right] \leq \sum_{\tau \in [2wd]} \Pr[B_\tau] \\ &\leq 2wd \max_{\tau \in [2wd]} \Pr[B_\tau] \end{aligned}$$

□

**Definition 1.15** (Noise Channel and The Local Stochastic Noise Property). A *noise channel*  $\mathcal{N}_\mathcal{E}$  is a distribution over sets of faults, we will denote the outcome of  $\mathcal{N}_\mathcal{E}$  by  $\mathcal{E}$ . For  $p \in (0, 1)$  we say that a noise channel  $\mathcal{N}_\mathcal{E}$  is a *local stochastic*, with noise rate  $p$ , if the following holds: For every subset of locations  $S \subset \mathbb{N} \times \mathbb{N}$ , whose locations have the same time coordinate, the probability that  $S$  is covered by  $\mathcal{E}$  is less than:

$$\Pr_{\mathcal{E} \sim \mathcal{N}_\mathcal{E}} [S \subset \mathcal{E}] \leq p^{|S|}$$

Observe that a quantum circuit  $C$  and a noise channel  $\mathcal{N}_\mathcal{E}$  induce a subjection  $C[\mathcal{E}]$ , which is a random variable.

**Claim 1.16.** Let  $\mathcal{N}_\mathcal{E}$  be a local stochastic noise channel with noise rate  $p$ , let  $C$  be a quantum circuit with bounded fan-in  $\Delta$ , and let  $c$  be an integer. We define  $q$  to be the number that depends on  $p$ ,  $\Delta$ , and  $c$  as:  $q := (\Delta^c p)^{\frac{1}{\Delta^c}}$ . There is a local stochastic noise channel  $\mathcal{N}_{\mathcal{E}_q}$ , with noise rate  $q$  that induces a subjection  $C[\mathcal{E}_q]_{\frac{1}{c}}$  such that:

$$C[\mathcal{E}]_{\frac{1}{2}} \sim_P C[\mathcal{E}_q]_{\frac{1}{c}}$$

*Proof.* We need to show that any subjection outcome  $C[\mathcal{E}]$  can be transformed by propagation to a circuit  $C'$  that is also a result of subjecting  $C$  to a set of faults  $\mathcal{E}'$  at frequency  $\frac{1}{c}$ , and that  $\mathcal{E}'$  satisfies the local noise property with a noise rate  $q$ . Consider the

□

For that we need the following property, for every subset of qubits  $S$ , there is subset of qubits  $S'$  at size at least  $(1 - \gamma)|S|$  such that for  $S$  to be deviated, all the qubits in  $S'$  have to absorb fault. Now suppose that the gate is at depth at most  $\Delta$ , and the fanning is at most  $c_f$ , then we get:

$$\begin{aligned} \Pr[S \in \mathcal{D}] &\leq (c_f^\Delta p)^{|S'|} \leq (c_f^\Delta p)^{(1-\gamma)|S|} \\ \Rightarrow p &\rightarrow (c_f^\Delta p)^{1-\gamma} \end{aligned}$$

A trivial lower bound for  $1 - \gamma$  is .